

Equivalence classes of integers under congruence modulo 4 and 5

$\text{mod } 4$	$\text{mod } 5$
$0 : \{\dots, -12, -8, -4, 0, 4, 8, 12, \dots\}$	$0 : \{\dots, -15, -10, -5, 0, 5, 10, 15, \dots\}$
$1 : \{\dots, -13, -9, -5, -1, 1, 5, 9, 13, \dots\}$	$1 : \{\dots, -16, -11, -6, -1, 1, 6, 11, \dots\}$
$2 : \{\dots, -14, -10, -6, -2, 2, 6, 10, 14, \dots\}$	$2 : \{\dots, -17, -12, -7, -2, 2, 7, 12, \dots\}$
$3 : \{\dots, -15, -11, -7, -3, 3, 7, 11, \dots\}$	$3 : \{\dots, -18, -13, -8, -3, 3, 8, 13, \dots\}$
	$4 : \{\dots, -19, -14, -9, -4, 4, 9, 14, \dots\}$

Say a, b, c, d are integers and n is a natural number.

If a is congruent to b modulo n ($a \equiv b \pmod{n}$) and c is congruent to d modulo n ($c \equiv d \pmod{n}$) do you think $a+c$ is congruent to $b+d$ modulo n ? Why or why not? (You can either choose to just create some examples or attempt a proof, it's up to you!)

Proof :

$a \equiv b \pmod{n}$ is the same as $a - b = k$ where k is some integer divisible by n .

$c \equiv d \pmod{n}$ is the same as $c - d = l$ where k is some integer divisible by n .

This means $n \mid k \wedge n \mid l$.

$$n \mid (k + l)$$

$$n \mid ((a - b) + (c - d))$$

$$n \mid (a + c) - (b + d)$$

This is the same as $a + c \equiv b + d \pmod{n}$.

Thus, the answer to the question is yes.